Real Time Recognition of Indian Sign Language in Emergency Situations using Hybrid CNN-RNN Architecture

J Kulandai Josephine Julina
Assistant Professor
josephinejulinajk@ssn.edu.in

Pavithra Manikandan
Student
pavithra2010140@ssn.edu.in

Gunanicaa Arun
Student
gunanicaa2010257@ssn.edu.in

Department of Information Technology
Sri Sivasubramaniya Nadar College of Engineering

***Abstract*- The comparison of various machine learning models for the recognition of the Indian Sign Language (ISL) is carried out in real time. The experiment clearly shows the effectiveness of each model namely Convolution Neural Network (CNN), Recurrent Neural Network (RNN), hybrid CNN-RNN, CNN-with the use of Google's MediaPipe and YOLOv8. ISL, crucial for communication within the hearing-impaired community, lacks widespread recognition systems, especially in critical situations. It emphasizes the societal significance of making ISL recognition more prevalent, particularly in critical infrastructure and emergency services. By breaking communication barriers, the comparison of the various technologies aims to enhance inclusivity and facilitate swift responses in high-stakes scenarios. The hybrid CNN-RNN architecture with the use of MediaPipe presented here stands as a promising solution for real-time ISL recognition, contributing to a more inclusive society prioritizing effective communication for all.**

***Index Terms* - CNN, RNN, YOLOv8, Indian Sign Language, ISL, prediction, opencv, tensorflow, mediapipe , INCLUDE, INCLUDE50.**

I. INTRODUCTION

In today's world, effective communication is often taken for granted, individuals with hearing and speaking impairments face unique challenges. One crucial means of communication is sign language which connects them with the world around them. The technological advancements in the field of sign language detection of the Indian Sign Language (ISL) offers a promising avenue to bridge the communications gaps.

The novel idea explored in this context is the combination of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) in a hybrid architecture with

Mediapipe and comparing this with other existing models like YOLOv8, a simple CNN. Here, both spatial as well as temporal features will be integrated in the recognition process. The CNN component effectively extracts the intricate and minute details of hand gestures and expression. Simultaneously, the RNN component takes care of the temporal dynamics inherent in sign language, hence ensuring a holistic understanding of the sequential nature of gestures.

This potential research will be a breakthrough for individuals with hearing and speaking impairments. Real-time recognition holds the promise of immediate and accurate interpretation of sign language, facilitating seamless communication in various scenarios. Irrespective of the scenario, this technology can empower individuals to express themselves more effectively and engage with the broader community.

This project plays an important role as it is a step towards being inclusive of all people. This will be achieved by the capabilities of deep learning to comprehend both the spatial intricacies and temporal nuances of sign language. In this pursuit, the comparison between various machine learning models is a crucial step in paving the way for a more inclusive and interconnected society.

II. LITERATURE REVIEW

The literature review encompasses several notable studies on Indian Sign Language (ISL) recognition, each contributing valuable insights to the field. In the early exploration [1] from 2014, the focus was on detecting numerical signs in ISL. The research involved creating a database with 5000 numeric sign images, utilizing the kNN clustering algorithm for classification and recognition. This

foundational work laid the groundwork for subsequent advancements. In a 2015 study [4], efforts were directed towards the recognition of alphabetic characters in ISL. The images underwent a transformation from RGB scale to YIQ and YUV space for skin segmentation. Despite experimenting with various machine learning models on the extracted features, the reported accuracies did not surpass 70%.

Moving forward to 2019 [2], real-time recognition of ISL incorporated the significance of facial recognition and hand gestures. The identification of the Region of Interest was achieved through the skin segmentation feature of OpenCV, and a Fuzzy C-means clustering algorithm was employed for gesture prediction.

In 2021, a study [3] specifically targeted alphabet recognition in ISL. The approach involved capturing hand gesture images through a webcam and applying computer vision techniques such as grayscale conversion, dilation, and mask operations. Convolutional Neural Networks (CNNs) were utilized for model training, showcasing the efficacy of deep learning in ISL recognition.

Further investigations in 2021 [7] explored the detection of ISL alphabets using machine learning algorithms, with the Random Forest algorithm achieving the highest accuracy for the images considered. Another study [6] evaluated deep learning models and optimizers, comparing pre-trained architectures like InceptionV3, ResNet152V2, and InceptionResNetV2. This research proposed a 3-layered CNN model, indicating a shift towards more intricate neural network architectures for ISL recognition based on images.

Furthermore, in 2020, a significant contribution was made in creating the Indian Lexicon Sign Language Dataset (INCLUDE). Until now, there was no reliable dataset for the Indian sign language, hence making this contribution very crucial. This provides a comprehensive collection of 0.27 million frames across 4,287 videos including 263 word signs from 15 different word categories. This dataset was recorded with the help of trained people hence ensuring the accuracy of the signs communicated. To help in faster evaluation, 50 words signs were selected across various categories from the INCLUDE dataset to create the INCLUDE-50 dataset. The authors of this contribution also evaluated several deep neural networks and recorded its accuracy. The best-performing model achieved an accuracy of 94.5% on the INCLUDE-50 dataset and 85.6% on the INCLUDE dataset. The authors further explored generalization by fine-tuning the decoder for an American Sign Language dataset, demonstrating an accuracy of 92.1% on the American Sign Language Lexicon Video Dataset(ASLLVD) with 48 classes. However, the research did not present any insights on real-time detection and recognition systems. Nonetheless, this research has provided a reliable dataset and insights into various deep neural network models to support the Indian sign language recognition, thus bridging crucial communication gaps.[8]

In 2020, another research was conducted on an approach for ISL alphabet recognition. The authors of this paper propose a computer vision-based system for recognizing Indian Sign Language (ISL) alphabets, aiming to provide an automatic and cost-effective solution to assist communication. The proposed system encompasses phases including image acquisition, preprocessing, segmentation, feature extraction, and classification. This was tested on all the 26 ISL alphabets and achieved an average accuracy of 80.76% and 100% accuracy when excluding similar alphabets from the testing dataset. However, this research was done only on the alphabets and not words or phrases of the Indian sign language, hence allowing for further advancements.[9]

The current landscape of Indian Sign Language (ISL) recognition systems reveals a focused exploration on the detection of numericals, characters, and alphabets. However, it is to be noted that the existing studies primarily rely on simulations rather than real-time applications. The absence of a real-time testing environment and the lack of consideration for sequence and temporal features are notable limitations. Furthermore, the focus on classification or identification from static images rather than dynamic video datasets is a prevailing trend. Moreover, further research is required in order to bring in an effective system for the recognition of the Indian Sign Language.

III. DATASET USED

The primary dataset used is the Indian Lexicon Sign Language Dataset (INCLUDE) which helps to focus on real-time recognition of Indian Sign Language (ISL). INCLUDE fills a significant void in Sign Language Recognition (SLR) research, as is the first publicly available dataset for ISL. The videos in this dataset have been recorded under the guidance of experienced signers and with input from the deaf community and hence is a more reliable and authentic dataset. The main advantages of this dataset is that it is open sourced.

The two principles that this dataset ensured to adhere to are authenticity and diversity. The videos were captured in

real-life scenarios and included a diverse array of signs and hence these principles were an integral part. The data collection process was recorded at a school for the deaf, with the help of seven senior students proficient in ISL. Multiple signers demonstrated each class or sign, with each signer contributing between 2 to 6 videos per class. In the recording, subjects were positioned facing the camera at a distance of 5-7 feet. Videos were filmed under natural lighting conditions, with no restrictions imposed on signer attire or signing style. All the recordings were conducted in a classroom setting.

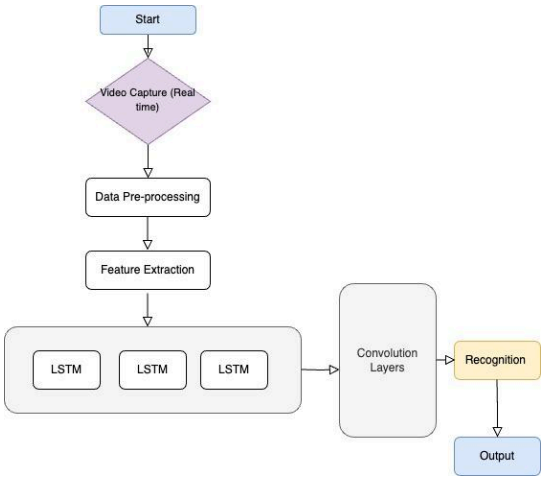
INCLUDE comprises 4,287 videos spanning 263 classes, where each class represents a distinct ISL sign. These classes are categorized into 15 broad categories, encompassing commonly used words in ISL, including adjectives, animals, greetings, occupations, and more. This dataset is characterized by short video durations, primarily ranging from 2 to 4 seconds, with an average duration of 2.57 seconds per sign. Each video maintains a resolution of 1920x1080 pixels and a frame rate of 25 fps, resulting in a total of 0.27 million frames encapsulating 263 unique signs.

The INCLUDE dataset is very useful for studying Indian Sign Language. Its open-source nature fosters transparency and accessibility. From this dataset, words related to emergency situations have been manually picked to ensure that communication for individuals with hearing and speaking impairments are enhanced in critical situations.

For the purpose of this research, 7 words have been picked which are alive, dead, loud, fast, healthy, sick and mother. Additionally, the dataset was manually labeled for training purposes. And hence this ensured that the data has been accurately labeled with minimal human error. For splitting the dataset, the standard principle of allocating 80% of the data for training and 20% for testing was used. This split was applied to each word in the dataset, hence maintaining uniformity across all categories. Therefore by using this dataset, the models have trained and tested, giving a reliable analysis of various models for the ISL recognition.

IV. PROPOSED SYSTEM

The proposed machine learning model that best worked for recognition of Indian Sign Language with highest accuracy is a hybrid CNN-RNN architecture using MediaPipe. In this paper, comparisons are drawn between different deep learning models such as CNN, hybrid CNN+RNN model, CNN+RNN along with mediapipe for implementing computer vision interference, and YOLOv8 trained on the above mentioned dataset. In the following subsections, the various models and their performance in terms of accuracy of training and testing data are presented.



A. *CNN*

A convolution neural network is a type of artificial neural network which is commonly used for image processing and classification tasks. In this research, CNNs were employed to recognise the hand gestures. Data loading and parsing of the XML label files are initial steps in this implementation. The CNN model is then defined using the Keras Sequential API which involves various convolutional layers followed by max pooling layers for feature extraction and fully connected layers for classification. The model was then compiled with the Adam optimizer and categorical cross-entropy loss function.

Parameters	**Settings**
Batch Size	32
Epochs	15
Conv2D_1 Filter (Filter Size)	32 (3x3)
MaxPooling2D_1	2x2
Conv2D_2 Filter (Filter Size)	64 (3x3)
MaxPooling2D_2	2x2
Conv2D_3 Filter (Filter Size)	128 (3x3)
MaxPooling2D_3	2x2
Dense_1	128
Output layer activation function	Softmax

B. *Hybrid CNN-RNN architecture*

As the next implementation model, a hybrid architecture merging Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) was adopted. This ensures the extraction of spatial features from individual frames with the help of the CNN and the extraction of temporal dependencies between these frames with the help of RNN. This hybrid approach is particularly advantageous for this task as understanding both spatial and temporal aspects is important. RNNs, specializing in sequential data processing, are very useful in capturing temporal dynamics and hence enhancing the model's ability to comprehend the flow of gestures over time.

Furthermore, the adaptability of RNNs accommodates the variations in signing styles and timing often encountered in ISL communication which adds to the advantages of this model. This holistic approach enhances the model's accuracy and reliability, facilitating the development of efficient ISL recognition systems tailored for emergency situations.

In this model, skin segmentation is used for the identification of the hands. This analyzes the color information to distinguish skin pixels from the background with the help of the set threshold values. This method helps focus the analysis on the hands, which are the primary element in Indian sign language communication. Once the hands are detected, it is then classified by the Machine Learning model.




C. *CNN+RNN implementation using Mediapipe*

This model is an extension of the previous model, it employs CNN as well as RNN for gesture recognition using the MediaPipe library in Python. This library uses the *Hands* module for detecting the hand landmarks and then uses a combination of CNNs and RNNs for classification. The CNN is responsible for extracting the spatial features and the RNNs for the temporal dependencies. This enables the model to effectively analyze and recognize the hand gestures.




This proposed model uses the softmax function as the final step to convert the raw output scores into a probability distribution over multiple classes:

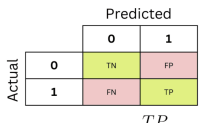
$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

The precision and recall values of the trained model are shown in the table below:

Class	Precision	Recall	F1 score
alive	0.929	0.902	0.915
dead	0.98	1	0.989
fast	0.987	0.9	0.941
healthy	0.9	0.96	0.93
loud	0.86	0.892	0.875
mother	0.94	0.913	0.926
sick	0.955	0.89	0.921

The images below represent how the precision, recall and F1 score values are calculated:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$



$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

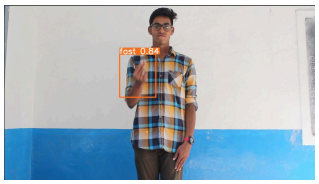
D. *YOLOv8*

YOLOv8, or You Only Look Once version 8, is an object detection model known for its speed and accuracy in

real-time applications. YOLOv8 divides the input image into a grid and directly predicts bounding boxes and class probabilities for each grid cell. It is an evolution of the YOLO family of models, which are designed to detect objects in images or video frames with high efficiency.

This model was trained on the dataset which contained annotated Indian Sign Language gestures. YOLOv8 can learn to detect the presence of these gestures and accurately localize them within the input images or frames.





Class	Precision	Recall	mAP50	mAP50-95
all	0.829	0.852	0.84	0.647
alive	0.878	0.902	0.767	0.6
dead	0.737	1	0.895	0.717
fast	0.9	0.607	0.788	0.679
healthy	0.815	0.75	0.556	0.479
loud	0.873	0.71	0.695	0.563

The loss function for the YOLOv8 model is simply the squared sum but modified to avoid weight localisation error. In addition to this, the modification of the loss function also avoids overpowering of a single grid cell's confidence score. The issues are solved by using λcoord and λnoobj. The loss function is shown below:

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left(C_i - \hat{C}_i \right)^2$$

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left(C_i - \hat{C}_i \right)^2$$

$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

V. PERFORMANCE METRICS

Accuracy, Precision, Recall, and F1 score are important ways to check how well a machine learning model can recognize Indian Sign Language (ISL) in emergencies. Accuracy shows how often the model gets the signs right out of all the signs it sees. Precision shows if the model gets the signs it says are correct, without making many mistakes. Recall helps to see if the model can find all the important signs without missing any. The F1 score is a mix of Precision and Recall, giving an overall idea of how good the model is. It must also be ensured that the model works quickly and doesn't use too much computer power, hence its computational efficiency is checked. By looking at these values, the efficiency of the model can be analyzed.

VI. RESULTS AND EVALUATION

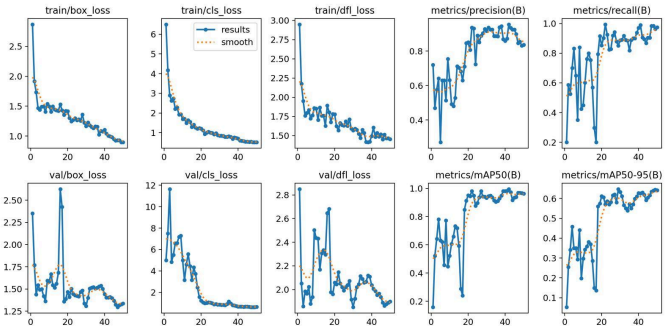
Below in this section is the individual test results for the various models that have been implemented. For all the models, the same 7 words relating to critical scenarios have been used.

The CNN model achieved an accuracy of 0.63, suggesting moderate performance in ISL recognition. However, with a loss of 0.66, indicating relatively high error, transition to the RNN+CNN implementation was made as the accuracy and robustness was to be improved. This shift aims to enhance recognition capabilities and support facilitating communication for the deaf and hard-of-hearing community.

The RNN+CNN model exhibited significant improvement over the previous model, achieving an accuracy of 0.8 and reducing the loss to 0.42. This enhanced performance shows the effectiveness of implementing recurrent neural networks alongside convolutional layers in ISL recognition. The model's higher accuracy and reduced loss signify enhanced precision and robustness, highlighting its potential to foster more seamless communication and inclusivity for individuals who are deaf or hard of hearing. To improve the accuracy further, the Mediapipe framework was used.

Transitioning from the RNN+CNN model the next algorithm implemented is YOLOv8. Yolov8 has been one of the efficient, but not the best, algorithms for the recognition of the Indian sign language. Its efficient object detection capabilities, YOLOv8 offers a unique approach to ISL recognition. With a precision score of 0.829, YOLOv8 demonstrates a high level of accuracy in correctly identifying ISL gestures, minimizing false positives and

ensuring reliable recognition.



The training and validation loss graphs offer visualizations of YOLOv8's learning progress over epochs.

Furthermore, it also achieves a recall score of 0.852, showing us that its ability to effectively capture a significant proportion of relevant ISL gestures from the dataset. This parameter is important for ensuring that the communication system is robust where every gesture carries essential meaning especially in critical scenarios.

Even though the mean average precision at 50% overlap (mAP50) score is at the value of 0.84, the mAP50-95 score is 0.647. This shows that when there is a moderate overlap between the predicted and ground truth bounding boxes, the YOLOv8 models performs well in terms of precisely locating the ISL gestures, but when it comes to the precision calculated across a range of IoU thresholds from 50% to 95%, it indicates that the model is facing challenges in maintaining consister precision across a wider range of IoU thresholds. IoU refers to Intersection over Union, which is a metric used to evaluate the accuracy of object detection algorithms. Both the mAP50 and mAP50-95 scores give us a broader perspective to analyze the models performance.

Returning to the proposed solution, the implementation of the CNN+RNN model along with the use of Mediapipe has proved to be the most efficient for recognizing Indian Sign Language gestures. With an overall accuracy score of 0.97 and loss of 0.1, the model exhibits robustness in classifying ISL gestures into seven distinct categories. The high precision, recall and f1-score scores proves that this method has been the most precise for detection of hand gestures.

Mediapipe played a crucial role in enhancing the effectiveness of the developed algorithm for ISL recognition in several ways. Firstly, Mediapipe provides a robust framework for hand tracking and landmark detection, which helps in precise localization of key points on the hand and fingers.

Moreover, Mediapipe's real-time performance capabilities allowed the algorithm to work efficiently. The ease of integration of Mediapipe also ensured seamless inclusion of the hand tracking and landmark detection functionality into the overall system. This integration streamlined the development process, hence there was more focus on fine-tuning the classification model and optimizing its performance.

Notably, the model achieves perfect precision and recall for certain classes, indicating its capability to accurately identify those gestures without false positives or negatives.

Overall, Mediapipe's robustness, real-time performance, and ease of integration significantly contributed to making the proposed and developed algorithm for ISL recognition the most efficient solution. By making use of Mediapipe, the algorithm achieved high accuracy and efficiency, paving the way for more effective communication and inclusivity for the deaf and hard-of-hearing community.

These impressive results underscore the potential of machine learning models to significantly contribute to the advancement of ISL recognition technology. This model lays a strong foundation for real-world applications in facilitating communication for the deaf and hard-of-hearing community and it is also suitable for deployment in real world scenarios.

VII. FUTURE WORK

Building upon this research, future work could delve deeper into fine-tuning and optimizing the machine learning models for real-time Indian Sign Language (ISL) recognition. Specifically, techniques such as hyperparameter tuning, architecture modifications, and dataset augmentation to further improve the performance metrics of the models can be studied. Nonetheless, one of the major advancements would be to investigate the feasibility of deploying these models on usable devices or platforms thereby making a more societal impact.

This research was constrained to a small portion of the entire INCLUDE dataset, hence implementation of these models for the entire dataset could also be one of the major improvements from this research.

Moreover, the integration of advanced deep learning architectures with multimodal inputs could be explored.

This can be done by combining visual information from images or video frames with additional modalities such as depth data from 3D sensors or temporal data from motion sensors. This system will help us adapt to diverse environments and user interactions. Moreover, leveraging techniques from transfer learning and domain adaptation will help in the deployment of sign language recognition systems in real-world settings with limited labeled data.

Another emerging avenue for future research lies in exploring the integration of sign language recognition technologies with augmented reality (AR) and virtual reality (VR) platforms. Building the user's field of view in AR or creating more interactive environments in VR, could enhance the effectiveness and inclusivity for the hearing-impaired community. Further exploration can include bidirectional communication between sign language users and non-signers by translating sign language gestures into text or spoken language, and vice versa using NLP techniques. Furthermore, collaborating with stakeholders from the community itself could ensure that the design and deployment of ISL recognition technologies are culturally sensitive, user-friendly, and aligned with the diverse needs and preferences of the end-users. This convergence of sign language recognition, with these technologies holds promise for bridging communication barriers and fostering greater inclusion in diverse social and educational contexts.

VIII. CONCLUSION

The importance of machine learning in advancing the recognition of Indian Sign Language for the deaf and hard of hearing community has been emphasized. The pivotal role of both developing robust recognition systems and raising awareness about Indian Sign Language (ISL) in India, where many individuals have limited knowledge of it has been underscored. Through the exploration and comparison of various models, particularly the hybrid CNN-RNN architecture with the integration of MediaPipe, demonstration of remarkable accuracy and efficiency in real-time ISL recognition has been made. The results signify a significant step towards breaking communication barriers and fostering inclusivity in critical scenarios, such as emergency services and infrastructure. By leveraging the capabilities of machine learning and computer vision technologies, enhancements have been made in not only communication accessibility but also exemplifies a commitment to creating a more inclusive and interconnected society. Moving forward, continued research and development in this field hold the promise of even greater advancements, ensuring that individuals with hearing and speaking impairments can fully participate in and contribute to society without communication barriers.

IX. REFERENCES

[1] H. Muthu Mariappan and V. Gomathi, "Real-Time Recognition of Indian Sign Language," 2019 International Conference on Computational Intelligence in Data Science (ICCIDS), Chennai, India, 2019, pp. 1-6, doi: 10.1109/ICCIDS.2019.8862125.

[2] Sahoo, Ashok. (2014). Indian sign language recognition using neural networks and kNN classifiers. Journal of Engineering and Applied Sciences. 9. 1255-1259.

[3] Prachi Sharma, Radhey Shyam Anand, A comprehensive evaluation of deep models and optimizers for Indian sign language recognition, Graphics and Visual Computing, Volume 5, 2021, 200032, ISSN 2666-6294

[4] Sharma, S., Singh, S. Recognition of Indian Sign Language (ISL) Using Deep Learning Model. Wireless Pers Commun 123, 671–692 (2022).

[5] M. Potnis, D. Raul and M. Inamdar, "Recognition of Indian Sign Language using Machine Learning Algorithms," 2021 8th International Conference on Signal Processing and Integrated Networks (SPIN), Noida, India, 2021, pp. 579-584, doi: 10.1109/SPIN52536.2021.9566141.

[6] Adithya V, Vinod PR, Gopalakrishnan U (2013) Artificial neural network based method for Indian sign language recognition. In: 2013 IEEE conference on information & communication technologies. IEEE, pp 1080–1085

[7] Choudhury A, Talukdar AK, Sarma KK (2014) A conditional random field based Indian sign language recognition system under complex background. In: 2014 fourth international conference on communication systems and network technologies. IEEE, pp 900–904

[8] Shenoy, Kartik & Dastane, Tejas & Rao, Varun & Vyavaharkar, Devendra. (2018). Real-time Indian Sign Language (ISL) Recognition. 1-9. 10.1109/ICCCNT.2018.8493808.

[9] Vashisth, Harsh Kumar, Tuhin Tarafder, Rehan Aziz, Mamta Arora, and Alpana. 2023. "Hand Gesture Recognition in Indian Sign Language Using Deep Learning" Engineering Proceedings 59, no. 1: 96. https://doi.org/10.3390/engproc2023059096